

Feeling Across the Distance: The Interplay of Motor and Empathic Processes with Virtual Humans

Scila Nunziata*

Renato Orti[†]

Francesco Ruotolo*

Gennaro Ruggiero[‡]Tina Iachini[§]

Laboratory of Cognitive Science and Immersive Virtual Reality, University of Campania Luigi Vanvitelli

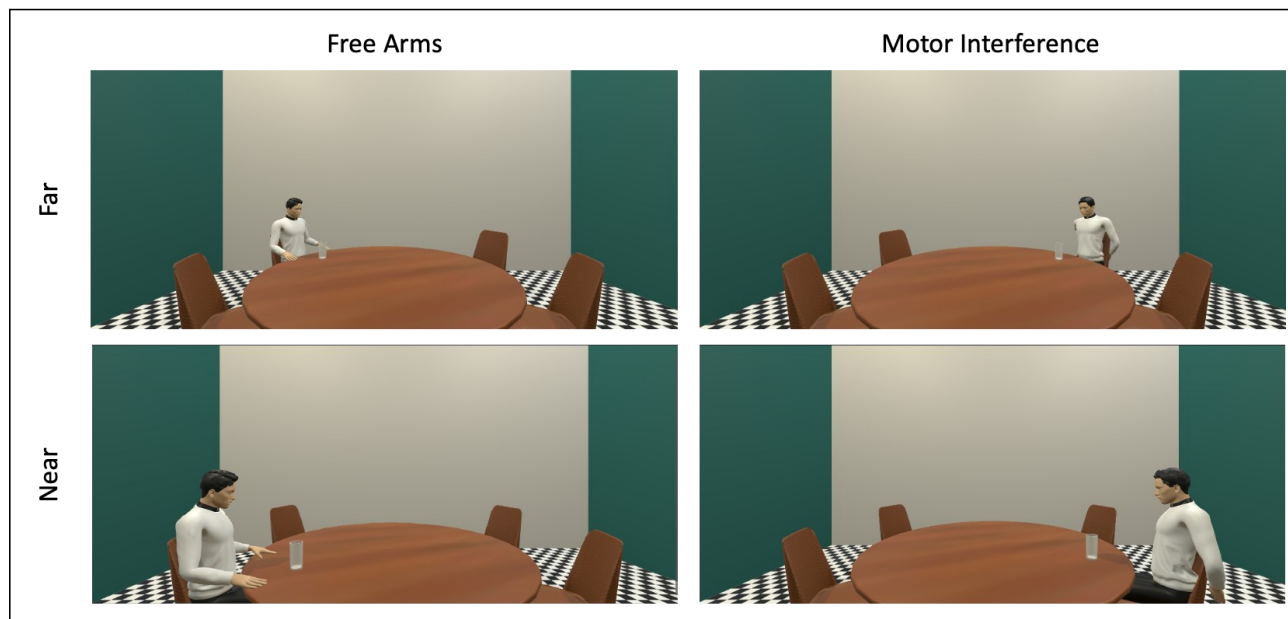


Figure 1. The figure shows, from the participant's point of view, the VH in the Near (bottom) and Far (top) portions of space, with his arms Free (left) or Blocked (right).

ABSTRACT

Immersive technologies enable increasingly frequent and complex interactions with "Virtual Humans" (VHs), now endowed with anthropomorphic behaviour and appearance. Therefore, it is important to explore the psychological mechanisms underlying social interaction between people and VHs. To this end, we investigated whether motor simulation mechanisms influence localization of stimuli at near (peripersonal) or far (extrapersonal) distances from a VH and whether empathic traits correlate with the task. In an immersive virtual reality environment, participants had to determine the position (right/left) of an object relative to VHs who could have their arms blocked (motor interference) or free (full motor availability) and could appear near or far from participants. Furthermore, we assessed the participants' empathic traits using the Interpersonal Reactivity Index (IRI). We hypothesized that motor interference should slow down the localization of objects in

peripersonal space, i.e., a space of action and defence. An association between empathic traits and spatial performance was expected, especially in extrapersonal space. The results confirmed these hypotheses, showing a deep engagement with the VHs based on motor simulation in peripersonal space and associated to the empathic tendency to identify with a fictitious character in extrapersonal space.

Index terms: Immersive Virtual Reality, Virtual Humans, Motor Simulation Mechanisms, Motor Interference, Near and Far Spaces, Empathic Traits.

1 INTRODUCTION

In the last few decades, the widespread use and advancement of immersive technologies have enabled increasingly frequent, complex and multi-purpose interactions between real humans and virtual humans (VHs), such as avatars and digital assistants. The inclusion of VHs makes the virtual experience more realistic, engaging and credible [1]. The human-VH interaction is thus becoming fundamental for various applications such as training, rehabilitation, therapy, and entertainment. As a result, we are often involved in "hybrid" (or quasi-social, see [2]) social relationships mediated by new simulative technologies with embedded VHs. This development calls for new research perspectives on how people perceive and mentally represent VHs, analysing their psychological and behavioural dynamics.

*e-mail: scila.nunziata@unicampania.it

[†]e-mail: renato.orti@unicampania.it

*e-mail: francesco.ruotolo@unicampania.it

[‡]e-mail: gennaro.ruggiero@unicampania.it

[§]e-mail: santa.iachini@unicampania.it

1.1 Prior work

Cognitive and psycho-social approaches have assessed some perceptual and behavioural features characterizing hybrid interactions. For instance, the anthropomorphic appearance, agency, gaze direction, age, gender and emotional expressions exhibited by VHs can influence the reactions of real people [3-8]. More specifically, these factors modulate people's "social" behaviour and their degree of "social presence", i.e. the feeling of being together with others with similar minds [9-10].

Past research has also shown that the modulation of distance between VHs and real people is a reliable index of the quality of hybrid interactions [13-14]. For example, we prefer a small distance from VHs when we feel safe and at ease or a large distance when we feel unsafe and uncomfortable [4,8,15]. Thus, external space is not homogeneous, but distances determine a fundamental spatial boundary between peripersonal (or near) and extrapersonal (or far) areas [16-17]. Peripersonal space can be defined as our sensorimotor interface with the environment, where we act in the here and now and react promptly to defend ourselves [18,19]. In this regard, Iachini and colleagues [20] asked participants to locate stimuli placed in front of VHs both from their first-person and by assuming the third-person perspective of VHs. The results revealed that it was more difficult to localise objects in peripersonal space, but not in extrapersonal space, if the VHs displayed blocked rather than free arms. This effect could reflect a motor simulation mechanism activated in near space, like what happens when we make judgements from our first-person perspective. This ability to put oneself in the other's shoes could be associated with empathy. Empathy is defined as a positive emotional capacity that involves understanding and adopting another person's perspective, as well as emotionally responding to their experiences [21]. However, although a study on proxemics has suggested that individual empathic predispositions may play a role in the interaction with VHs [11], this factor has not been explored to date.

1.2 The current study

We aimed at verifying whether our interaction with VHs is affected by the interplay between spatial boundaries, motor availability and individual empathic traits. To this end, participants had to determine the position (right/left) of an object (a glass) relative to a VH. VHs could have their arms blocked (motor interference) or free (full motor availability) and be located in near (peripersonal) or far (extrapersonal) space. In addition, the Interpersonal Reactivity Index (IRI) questionnaire [21] was used to assess participants' empathic traits. In this way, we could assess the combined effect of these factors to answer the following research questions:

RQ1- does the position of the VHs and their motor availability influence the participants' response?

RQ2- is the response of participants influenced by individual empathic predispositions?

We hypothesized that:

H1- observing motor interference should disrupt the spatial location of objects if the VH was located at near but not far distance from the participant.

H2- empathic traits should be associated with the performance, especially in far space where motor resources are less relevant.

2 METHOD

2.1 Participants

Young healthy participants, with normal/corrected-to-normal vision and free from neurological disorders, were recruited for the study via advertising on social networks and word of mouth. Once recruited for the study, participants gave their written consent to take part in the study. All participants enrolled in the study declared

to be right-handed. The study was conducted in conformity with the local Ethics Committee requirements and the Declaration of Helsinki [22].

2.2 Setting and apparatus

The experiment took place in a sound proofed room (size $5 \times 4 \times 3$ m) of the Laboratory of Cognitive Science and Immersive Virtual Reality (Lab CS-IVR). The IVR equipment consisted of a Sony HMZ-T1 (SONY, JAP) head-mounted display (HMD) with two OLED displays providing stereoscopic depth with a screen resolution of 1280×720 pixel (per eye) and a refresh rate of 60 Hz. The virtual scenario spanned 45° horizontally and 51.6° diagonally. The IVR system allowed participants to experience dynamic stereoscopic visual stimuli in a naturalistic way.

2.3 Virtual Scenario and Virtual Stimuli

The virtual stimuli consisted of a three-dimensional room ($3 \times 2.4 \times 3$ m), displayed with green walls and chessboard-textured floor, containing a chair, a circular table, and a glass on it (Figure 1). The virtual stimuli were created by means of the 3D modelling free software SketchUp (Trimble, <https://www.sketchup.com>) and were imported into the Virtual Reality Software Toolkit (WorldViz LLC) similarly to previous studies [20, 23-24]. A young male adult VH was selected from a database of characters (Vizard Complete Characters, WorldViz, USA).

2.4 Interpersonal Interactivity Index (IRI)

The IRI questionnaire [21] measures on a 5-step Likert-type scale (from "never true" to "always true") various facets of empathy. More precisely, participants have to rate their empathic predispositions through four subscales, each comprising 7 items. The Perspective Taking subscale evaluates the tendency to adopt the psychological point of view of others (e.g., "I sometimes try to understand my friends better by imagining how things look from their perspective"). The Fantasy subscale assesses the tendency to identify with a fictional character (e.g., "After seeing a play or movie, I have felt as though I was one of the characters"). The Empathic Concern subscale evaluates the tendency to experience feelings of sympathy and compassion toward unfortunate others (e.g., "I often have tender, concerned feelings for people less fortunate than me"). Finally, the Personal Distress subscale regards the tendency to experience discomfort in distress situations (e.g., "Being in a tense emotional situation scares me").

2.5 Procedure

After completing the IRI questionnaire (RQ2), participants received written instruction about the task that were orally revised. Then, they were invited to sit on a comfortable chair, to wear the HMD and to put their left hand on a mouse. Participants were asked to use their non-dominant left hand to avoid feeling stuck if forced to keep their dominant hand fixed on the mouse during the object localisation task. Before starting the experimental session, participants underwent a training session aimed at clarifying the experimental procedure and making them familiar with the IVR equipment. If everything was clear, the experimental session started.

In the virtual room, the participant saw the circular table, the VHs and the glass in front of them (see Figure 1). The participant sat in a chair in the laboratory corresponding to the virtual chair. The VHs could appear in four positions in relation to the participants: in the peripersonal sector of space, at 40 cm away on their left or right side (Near-Left, Near-Right), in the extrapersonal sector of space at 150 cm away on their left or right side (Far-Left, Far-Right). The VHs could appear in each position with their arms free (Free condition) or blocked behind their back (Blocked condition) (RQ1).

The glass always appeared close to the VHs, 20 cm from them. The glass could appear on the right or left side in relation to the midline of the VH's body-midline at four different distances on each side: 4 mm, 8 mm, 12 mm or 16 mm, as already used in previous studies [20]. Participants had to judge whether the glass appeared to the right or left of the VHs' body (left/right localization judgement). The glass could appear in eight right/left positions, for all Free and Blocked conditions (total = 64 trials). Each trial started with a grey fixation cross. Participants had to fixate the cross (100 ms), followed by a blank screen (1000 ms) and then stimuli were presented (300 ms). Once the stimulus disappeared, they had to locate the position of the glass by pressing the corresponding right or left mouse button. The participants were instructed to respond as accurately and quickly as possible. Nobody claimed discomfort or vertigo during the IVR experience or reported being aware of the experimental purpose.

Response time (RTs, in seconds) of correct responses measured participants' performances. The overall accuracy of the performance was quite high (>90%).

At the end of the experimental session, participants were asked to report whether they encountered difficulties in the task, whether the stimuli were easily discriminable, and whether or not they could touch the glass and the VH (on the use of verbal reports to assess the perceptual experience in IVR see [25]). Participants reported that they did not encounter any difficulties in the task, that VHs and glass were easily discriminable and reachable in nearby space and were easily discriminable but unreachable when they were far away.

2.6 Data analysis

Thirty-eight right-handed participants (11 males) aged between 20-37 years ($M_{age} = 25.58$; $SD_{age} = 4.32$), took part in the experiment.

The assumption of normality was checked by means of skewness and kurtosis: average skewness = 0.45, range: -0.51 to 0.96; average kurtosis = 0.50, range: -0.61 to 1.17 [26].

Participants' performance on spatial localization was analysed by means of a Repeated Measures ANOVA carried out on mean RTs, with Position as a 4-level within-subject variable (Near-Left, Near-Right, Far-Left, Far-Right) and Motor Interference as a 2-level within-subject variable (Free, Blocked) (RQ1). The Newman-Keuls test was used to analyse post-hoc effects, and the magnitude of the significant effects were indicated by means of partial eta squared (η_p^2).

The association of participants' empathic traits with spatial localization was assessed through Pearson's correlation analysis between mean RTs and mean scores of the four IRI subscales.

Finally, considering the results of the correlation, we performed separate linear regression analyses with the IRI subscale Fantasy as predictor and the four RTs of spatial localization in far space (four conditions) as criteria (RQ2).

3 RESULTS

3.1 ANOVA on Response Times (RTs)

The ANOVA showed a main effect of Position ($F(3, 111) = 3.774$, $p < 0.01$, $\eta_p^2 = 0.09$). Participants were slower in the Far-right ($M = 0.761$, $SD = 0.31$) than in the Near-Left ($M = 0.668$, $SD = 0.26$) and Near-Right ($M = 0.674$, $SD = 0.24$) conditions (at least $p < 0.05$). Furthermore, a significant main effect of Motor Interference appeared ($F(1, 37) = 6.830$, $p < 0.01$, $\eta_p^2 = 0.16$). Participants were faster in the Free ($M = 0.675$, $SD = 0.26$) than Blocked condition ($M = 0.728$, $SD = 0.29$) (at least $p < 0.01$). A significant interaction between Position and Motor Interference emerged ($F(3, 111) = 8.443$, $p < 0.001$, $\eta_p^2 = 0.19$). In Near space, participants were faster in all Free (Left $M = 0.598$, $SD = 0.20$; Right $M = 0.625$, $SD = 0.18$) compared to Blocked (Left $M = 0.737$, $SD = 0.29$; Right $M = 0.724$,

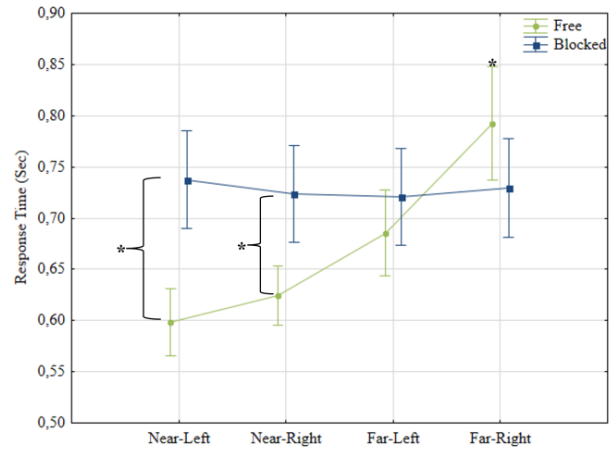


Figure 2. The graph shows the RTs in Near-Left, Near-Right, Far-Left, Far-Right positions as a function of the Free and Blocked conditions. Vertical bars represent standard errors. * $p < .05$

$SD = 0.28$) conditions. Such difference did not emerge in Far conditions ($p > 0.05$). Finally, when observing VHs with free hands, individuals were slower in the Far-Right ($M = 0.793$, $SD = 0.34$) than Far-Left ($M = 0.685$, $SD = 0.25$), Near-Right ($M = 0.625$, $SD = 0.18$) and Near-Left ($M = 0.598$, $SD = 0.20$) conditions (at least $p < 0.05$) (Figure 2).

3.2 Correlation between Response Times (RTs) and IRI Subscales

The results of Pearson's correlation analysis between mean RTs and mean IRI subscales are illustrated in Table 1. The results showed that the more the participants reported a tendency to identify with fictitious characters ("Fantasy" subscale of the IRI, [21]), the faster they were in locating the glass only in the extrapersonal sector of space.

Table 1. The table shows correlation coefficients (Pearson's r) between the mean reaction times (RTs) for spatial judgments in each of the eight conditions and the mean scores on each of the IRI subscales

Condition	Fantasy	Empathic Concern	Perspective Tacking	Personal Distress
Near-Left Free	-0.25	-0.22	0.08	-0.06
Near-Left Blocked	-0.22	-0.03	-0.13	-0.10
Near-Right Free	-0.30	-0.29	-0.11	-0.04
Near-Right Blocked	-0.29	-0.02	-0.14	-0.04
Far-Left Free	-0.44**	-0.09	0.00	-0.10
Far-Left Blocked	-0.41*	-0.12	0.02	-0.12
Far-Right Free	-0.38*	-0.14	-0.17	-0.10
Far-Right Blocked	-0.35*	-0.15	-0.13	-0.12

* $p < 0.05$, ** $p < 0.01$

3.3 Regression between Response Times (RTs) and IRI Subscales

The results showed that the IRI subscale "Fantasy" [21] predicted RTs of spatial localization in the four Far conditions: Left Free ($F(1,36) = 8.639, p < 0.05, R = -0.44, R^2 = 0.19$), Right Free ($F(1,36) = 5.919, p < 0.05, R = -0.38, R^2 = 0.14$), Left Blocked ($F(1,36) = 7.130, p < 0.05, R = -0.41, R^2 = 0.17$), and Right Blocked ($F(1,36) = 5.017, p < 0.05, R = -0.35, R^2 = 0.12$). Overall, as shown in Figure 3, the increase in participants' tendency to identify with fictitious characters led to a faster RT to localize the glass with respect to the VHs in all Far positions.

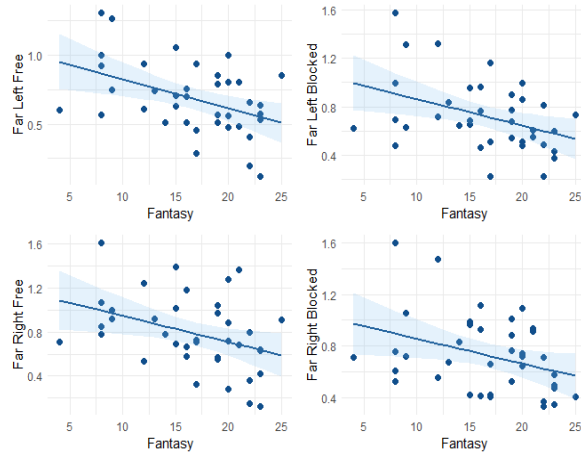


Figure 3. Scatterplots depicting the significant negative correlations between participants' scores on the "Fantasy" subscale of the IRI and their reaction times (RTs) for spatial judgments in each of the four conditions in the far space.

4 DISCUSSION

Nowadays, the opportunities to interact with VHs are increasing worldwide and this poses new challenges in understanding the psychological mechanisms underlying these social interactions. This study aimed to investigate the combined effect of spatial boundaries, motor availability and empathic traits on participants' interactions with VHs. To this end, participants were immersed in a virtual environment and had to locate a glass by assuming the perspective of VHs who could have their arms free or blocked behind their backs and could appear at near (peripersonal) or far (extrapersonal) distances. Participants' empathic characteristics were also assessed [21].

Overall, the results showed that participants were slower in localizing the objects when the VHs appeared at far than near distances, and when they had their arms blocked than free, as also reported by previous studies [20]. However, the crucial result was that localization was slower when participants saw VHs close to them with their arms blocked rather than free. In contrast, when the VHs were far away, motor interference had no effect. This finding confirms our hypothesis (H1). In this way we have demonstrated that in assuming the perspective of VHs we can also implicitly simulate their action possibilities, and this only happens when we share a common peripersonal space [20,24].

Interestingly, participants' localization performance was faster as the tendency to identify themselves with a fictitious character [20] increased. Notably, as expected (H2), the association between

participants' spatial performance and empathic disposition towards fictional characters emerged when VHs appeared at far but not near distances. This result was corroborated by the linear regression analyses demonstrating that higher scores on the Fantasy subscale predicted faster RTs only when VHs were in far distances, both with free and blocked arms. This could be interpreted as stemming from the fact that when the motor component loses relevance in the far space, individual empathic traits towards VHs acquire higher relevance. Instead, when VHs are in near space, factors related to the need to act and defend oneself in time prevail, and among these the first place goes to the full availability of motor resources. Importantly, this mechanism is demonstrated when we have our arms free and assume the perspective of a VH with blocked arms, similarly to when we have our arms blocked and locate stimuli close to our body in the first-person perspective [20, 24].

However, the current study presents some limitations. First, direct measures of the degree of presence (how much participants felt themselves to be in the virtual rather than physical environment) and/or social presence (how much participants felt to be with another person) [10, 28] were not included. Second, more objective measures of the interaction with VHs are missing. Future studies should complement behavioural and self-report measures with more objective data such as psychophysiological indexes.

5 CONCLUSION

In conclusion, the results reveal that the position of the VHs and their motor availability influence the participants' response. Besides, individual empathic predispositions are associated with the response of participants. However, their strength emerges when the primary spatial and motor mechanisms become less relevant, as they do when VHs are far from real individuals. Therefore, the relationship with VHs can become so deep that we put ourselves in their shoes by simulating their action possibilities, thus suggesting that this primordial adaptive mechanism [20,24] also works in virtual worlds.

Given the paramount importance of human-VH interaction across diverse applications like training, rehabilitation, therapy and entertainment, these findings hold significant value for the design of dedicated virtual environments. Specifically, the posture and spatial distance of VHs relative to the user are important to improve the credibility and realism of hybrid interactions, as necessary in the various applied fields.

6 REFERENCES

- [1] Frank Biocca, Kim, Jin and Yung Choi, "Visual touch in virtual environments: An exploratory study of presence, multimodal interfaces, and cross-modal sensory illusions", *Presence: Teleoperators & Virtual Environments*, MIT Direct Press, USA, pp. 247-265, 2001.
<https://doi.org/10.1162/105474601300343595>
- [2] Alex M. Andrew, "Virtual Reality: Exploring the Brave New Technologies of Artificial Experience and Interactive Worlds from Cyberspace to Teledildonics by Howard Rheingold Seeker and Warburg", *Robotica*, Cambridge University Press, London, 1991.
<https://doi.org/10.1017/S0263574700008067>
- [3] Jeremy N. Beilenson, Andrew Beall and Jim Blascovich, "Gaze and task performance in shared virtual environments", *The journal of visualization and computer animation*, John Wiley & Sons, USA, 2002.
<https://doi.org/10.1002/vis.297>
- [4] Jeremy N. Beilenson, Jim Blascovich, Andrew C. Beall and Jack M. Loomis, "Interpersonal distance in immersive virtual environments",

- Personality and Social Psychology Bulletin, SAGE Publications, USA, 819-833, 2003.
<https://doi.org/10.1177/0146167203029007002>
- [5] Timothy W. Bickmore and Rosalind W. Picard, "Establishing and maintaining long-term human-computer relationships", ACM Transactions on Computer-Human Interaction, ACM Digital Library, USA, pp. 293–327, 2005.
<https://doi.org/10.1145/1067860.1067867>
 - [6] Karl F. MacDorman, Robert D. Green, Chin-Chang Ho and Clinton T. Koch, "Too real for comfort? Uncanny responses to computer generated faces", Computers in Human Behaviour, Elsevier, Netherlands, pp. 695–710, 2009.
<https://doi.org/10.1016/j.chb.2008.12.026>
 - [7] Masahiro Mori, Karl F. MacDorman and Norri Kageki, "The uncanny valley [from the field]", IEEE Robotics & Automation Magazine, IEEE Xplore, USA, pp. 98–100, 2012.
<https://doi.org/10.1109/MRA.2012.2192811>
 - [8] Tina Iachini, Yann Coello, Francesca Frassinetti and Gennaro Ruggiero, "Body Space in Social Interactions: A Comparison of Reaching and Comfort Distance in Immersive Virtual Reality", PLoS ONE, USA 2014.
<https://doi.org/10.1371/journal.pone.0111511>
 - [9] Kristine L. Nowak and Frank Biocca, "The effect of the agency and anthropomorphism on users' sense of telepresence, copresence, and social presence in virtual environments", Presence: Teleoperators and Virtual Environments, MIT Direct Press, USA, pp. 481–494, 2003.
<http://dx.doi.org/10.1162/105474603322761289>
 - [10] Frank Biocca, Chad Harms and Judee K. Burgoon, "Towards a more robust theory and measure of social presence: Review and suggested criteria". Presence Teleoperator & Virtual Environments, MIT Direct Press, USA, pp. 456–480, 2003.
<http://dx.doi.org/10.1162/105474603322761270>
 - [11] Mariachiara Rapuano, Tina Iachini, Francesco Ruotolo, Alessandro Troise, Md Sheeraz Anwar and Gennaro Ruggiero, "The Role of Empathic Traits in the Interaction with Virtual Humans", 2022 IEEE International Conference on Metrology for Extended Reality, Artificial Intelligence and Neural Engineering, IEEE MetroXRaine, USA, pp. 352–356, 2022.
<https://doi.org/10.1109/MetroXRaine54828.2022.9967636>
 - [12] Christos Kyrilitsias and Despina Michael-Grigoriou, "Social interaction with agents and avatars in immersive virtual environments: A survey", Frontiers in Virtual Reality, Frontiers Research Foundation, Switzerland, 2022.
<https://doi.org/10.3389/frvir.2021.786665>
 - [13] Scila Nunziata, Antonella Ferrara, Tina Iachini, Renato Orti, Alessandro Troise and Gennaro Ruggiero, "Virtual Agents and Proxemic Distances: How Social Interactions Affect Our Spatial Representations", 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering, IEEE MetroXRaine, USA, pp. 922–927, 2023.
<https://doi.org/10.1109/MetroXRaine58569.2023.10405632>
 - [14] Gennaro Ruggiero, Francesca Frassinetti, Yann Coello, Mariachiara Rapuano, Armando S. di Cola and Tina Iachini, "The effect of facial expressions on peripersonal and interpersonal spaces", Psychological Research, Springer, Germany, pp. 1232–1240, 2017.
<https://doi.org/10.1007/s00426-016-0806-x>
 - [15] Ann Huang, Pascal Knierim, Francesco Chiossi, Lewis L. Chuang, Robin Welsch, "Proxemics for human-agent interaction in augmented reality", in Proceedings of the 2022 CHI conference on human factors in computing systems, ACM SIGCHI, USA, pp. 1–13, 2022.
<https://doi.org/10.1145/3491102.3517593>
 - [16] Anna Berti and Francesca Frassinetti, "When Far Becomes Near: Remapping of Space by Tool Use", Journal of Cognitive Neuroscience, MIT press, USA, pp. 415–420, 2000.
<https://doi.org/10.1162/089892900562237>
 - [17] Yann Coello and Tina Iachini, "Embodied perception of objects and people in space: Towards a unified theoretical framework". in Perceptual and emotional embodiment, Routledge, United Kingdom, pp. 206–228, 2015.
<https://doi.org/10.4324/9781315751979>
 - [18] Frederique de Vignemont and Giandomenico Iannetti, "How many peripersonal spaces?", Neuropsychologia, Elsevier, United Kingdom, pp. 327–334, 2015.
<https://doi.org/10.1016/j.neuropsychologia.2014.11.018>
 - [19] Gennaro Ruggiero, Mariachiara Rapuano, Alice Cartaud, Yann Coello and Tina Iachini, "Defensive functions provoke similar psychophysiological reactions in reaching and comfort spaces", Scientific Reports, Nature Portfolio, United Kingdom, 2021.
<https://doi.org/10.1038/s41598-021-83988-2>
 - [20] Tina Iachini and Gennaro Ruggiero, "Can I put myself in your shoes? Sharing peripersonal space reveals the simulation of the action possibilities of others", Experimental Brain Research, Springer Nature Link, Germany, pp. 1035–1045, 2021.
<https://doi.org/10.1007/s00221-021-06040-9>
 - [21] Mark H. Davis, "Interpersonal reactivity index", 1980.
<https://psycnet.apa.org/doi/10.1037/101093-000>
 - [22] W. M. Association, "World Medical Association Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Subjects", JAMA, USA, pp. 2191–2194, 2013.
 - [23] Tina Iachini, Gennaro Ruggiero, Francesco Ruotolo and Michela Vinciguerra, "Motor resources in peripersonal space are intrinsic to spatial encoding: Evidence from motor interference", Acta psychologica, Elsevier, Netherlands, 2014.
<https://doi.org/10.1371/journal.pone.0111511>
 - [24] Scila Nunziata, Renato Orti, Antonella Ferrara, Francesco Ruotolo, Gennaro Ruggiero and Tina Iachini, "We implicitly empathize with virtual agents: the effect of motor simulation", in 2022 IEEE International Conference on Metrology for Extended Reality, Artificial Intelligence and Neural Engineering (MetroXRaine) IEEE, USA, pp. 357–361, 2022.
<http://dx.doi.org/10.1109/MetroXRaine54828.2022.9967627>
 - [25] Jack M. Loomis and Joshua M. Knapp, "Visual perception of egocentric distance in real and virtual environments", In Virtual and adaptive environments, CRC Press, USA, pp. 21–46, 2003.
<https://psycnet.apa.org/doi/10.1201/9781410608888.pt1>
 - [26] Chih-Ping Chou and Peter M. Bentler. "Model modification in structural equation modelling by imposing constraints", Computational statistics & data analysis, Elsevier, Netherlands, pp. 271–287, 2002.
[https://doi.org/10.1016/S0167-9473\(02\)00097-X](https://doi.org/10.1016/S0167-9473(02)00097-X)
 - [27] Michel Graziano, "The spaces between us: A story of neuroscience, evolution, and human nature", Oxford University Press, UK, 2017.
 - [28] Maria V. Sanchez-Vives and Mel Slater, "From Presence to Consciousness through Virtual Reality", Nature Reviews Neuroscience, Nature Portfolio, pp. 332–339, 2005.
<https://doi.org/10.1038/nrn1651>